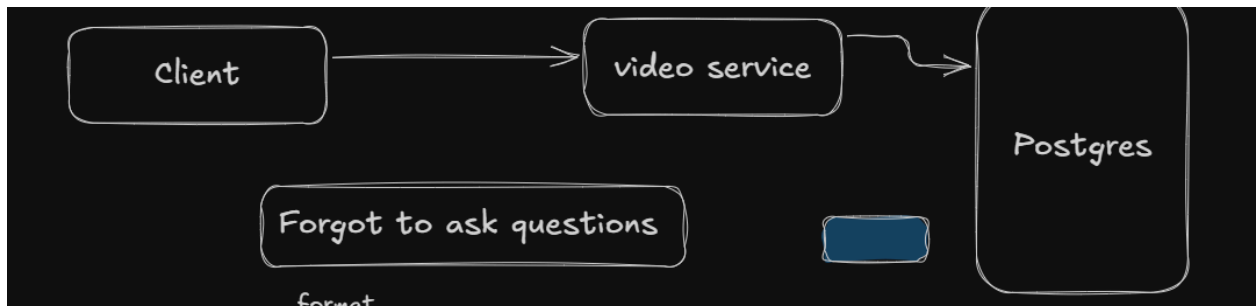


Build Video solution

1. Today Hitesh sir Build a complete video solution similar to YouTube, Vimeo, Learnyst, and Teachyst.
2. Two type of video
 - Streaming like hotstar
 - Video solution i.e. hosted video like youtube
3. So hitesh sir will focus on Hosted video system design
4. Users upload videos to the backend server, where the video files are stored in a PostgreSQL database as binary data using the **BYTEA** column type.



★ AI Overview 🔊

Bytea is a PostgreSQL data type used to store variable-length binary strings, or raw bytes (up to 1GB). Unlike text types, it allows storing non-printable bytes and zero-value octets, making it ideal for binary data like images, audio files, or encrypted data. It commonly uses hexadecimal (\x...) formatting for input/output. PostgreSQL +3

Key Aspects of bytea

- **Purpose:** Designed for storing binary data, not text, meaning it ignores character set

What are we missing here?

We forgot to ask the right questions. Without asking proper questions, we cannot design a good system.

Lets ask a question to the Interviewer?

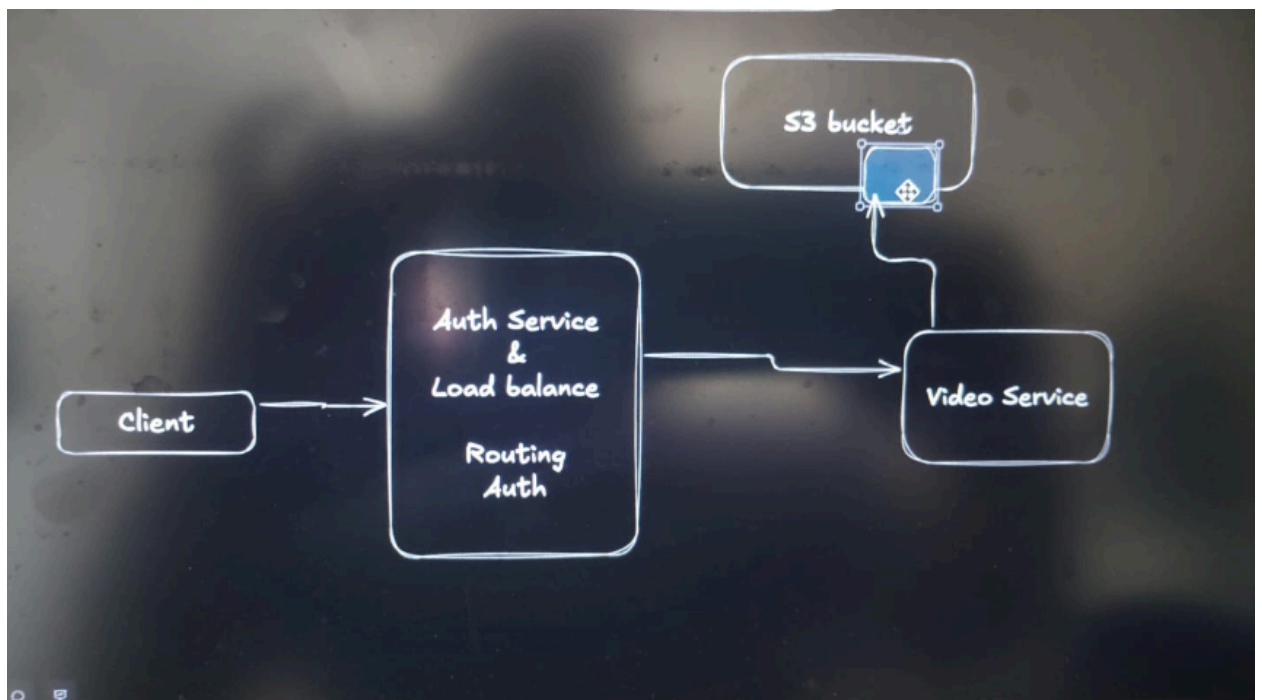
format
duration
size
number of daily upload
are we supporting various formats (encoding)

Auth, high video upload, fast viewing experince,

Video on Demand (VOD) feature: **Video on Demand (VOD)** is a system that allows users to watch video content anytime they want

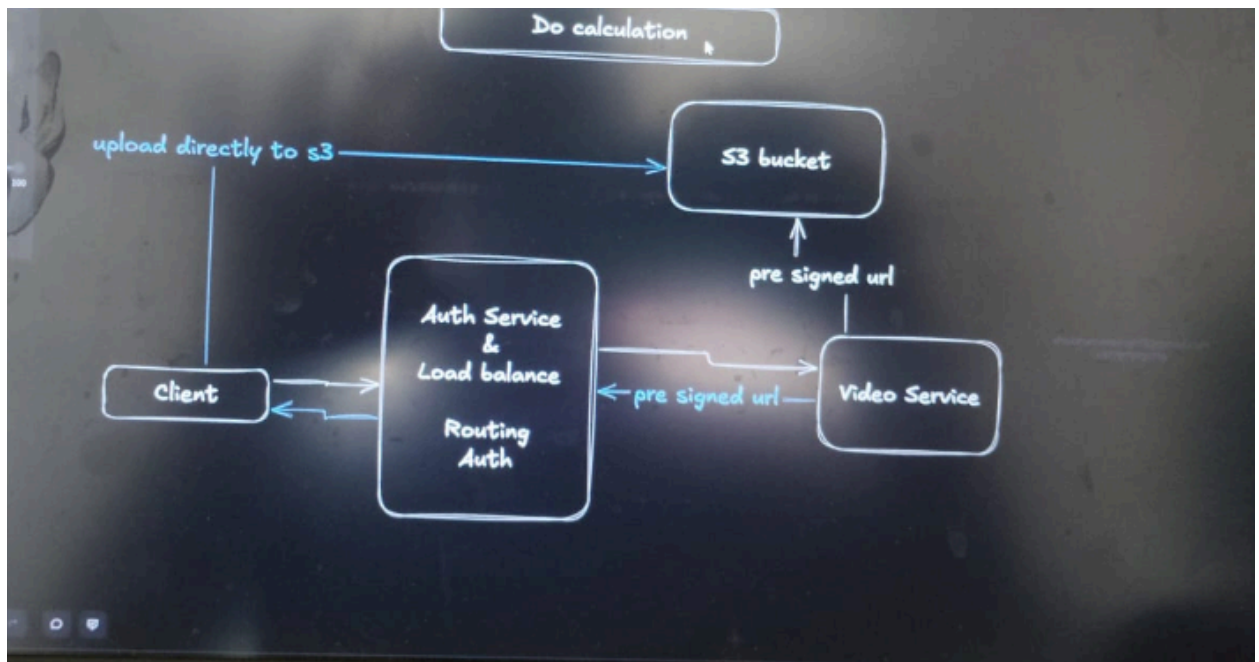
Note: Don't ask more feature because there is some limited time in system design interview

1.



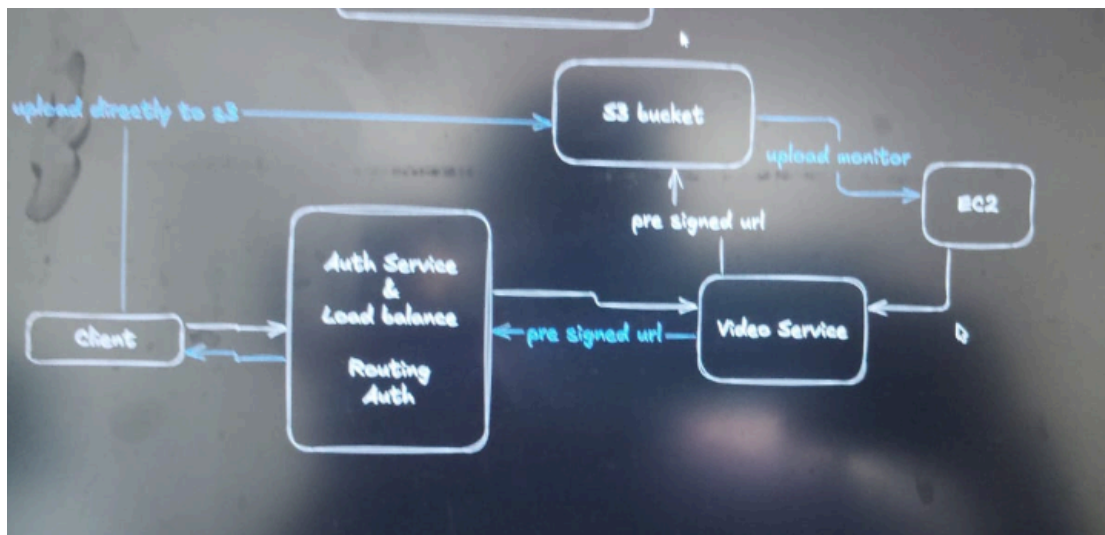
Here video service is Our Backend (EC2 machine/server)

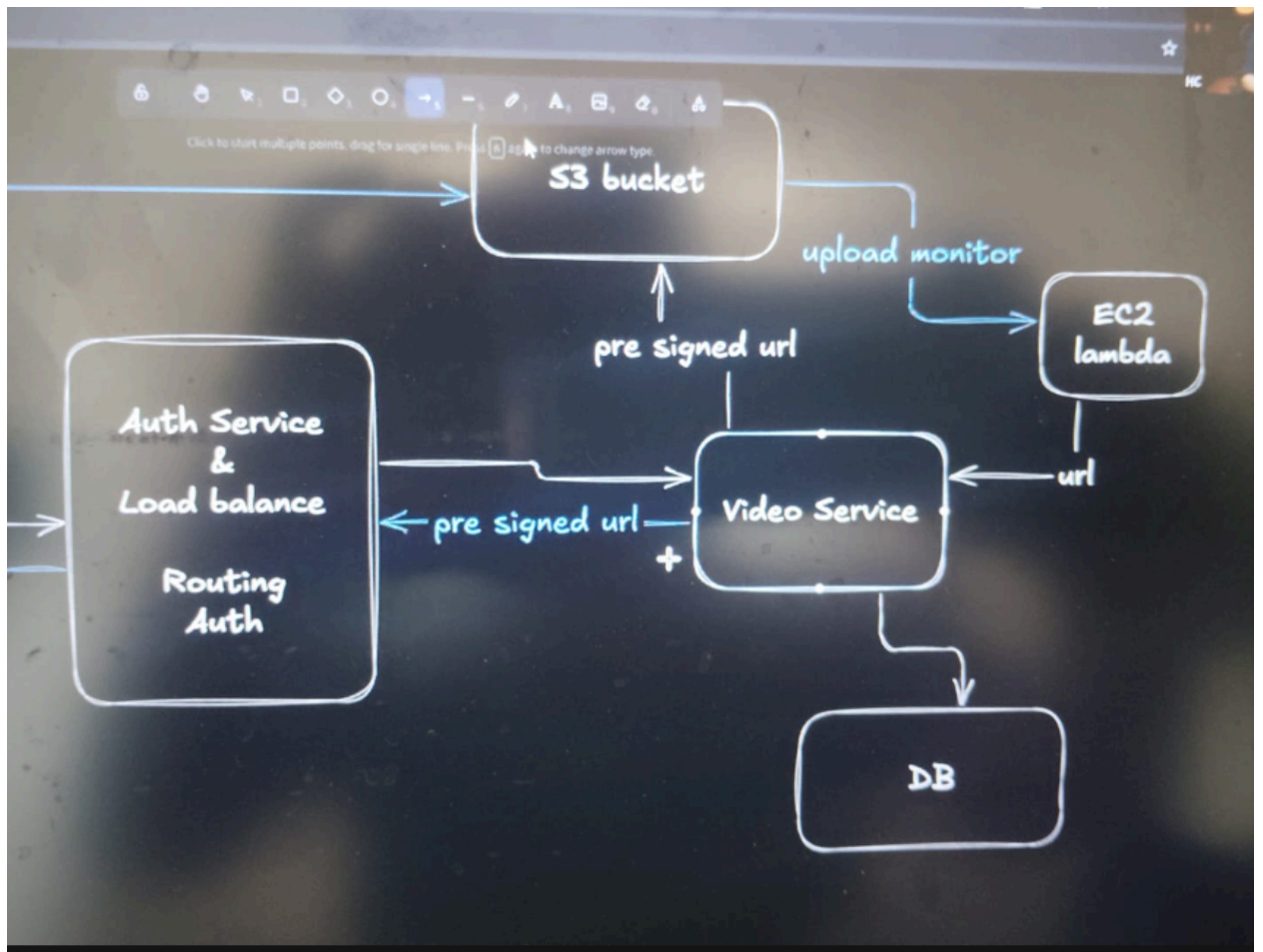
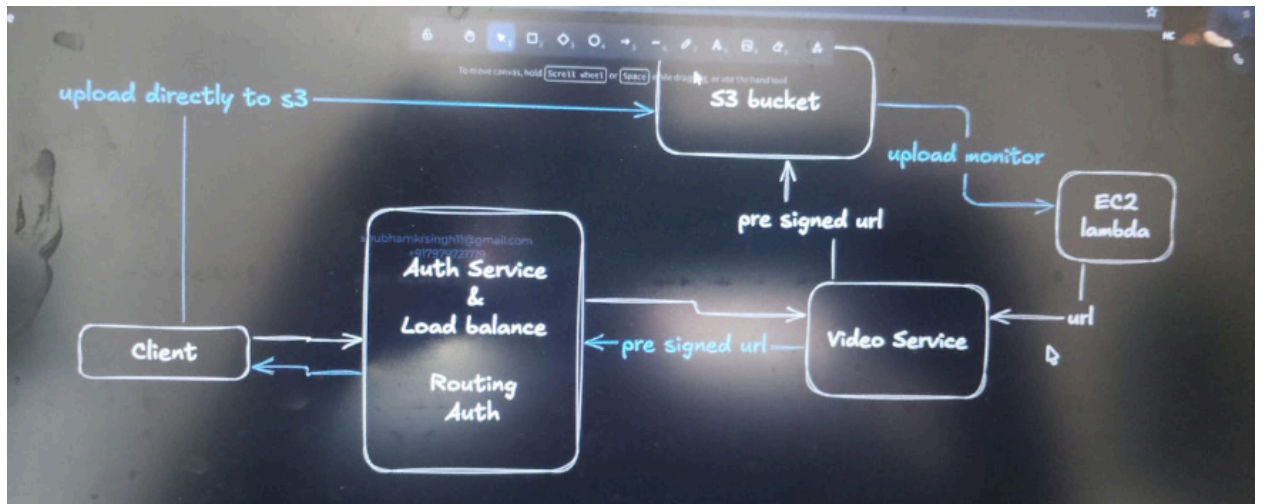
2.

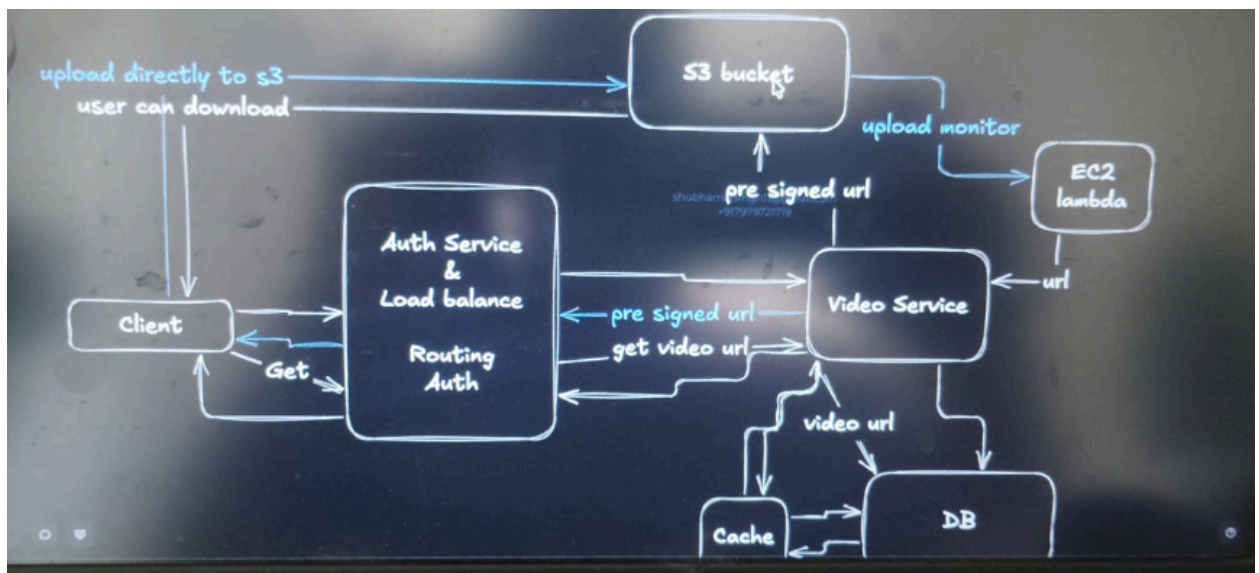
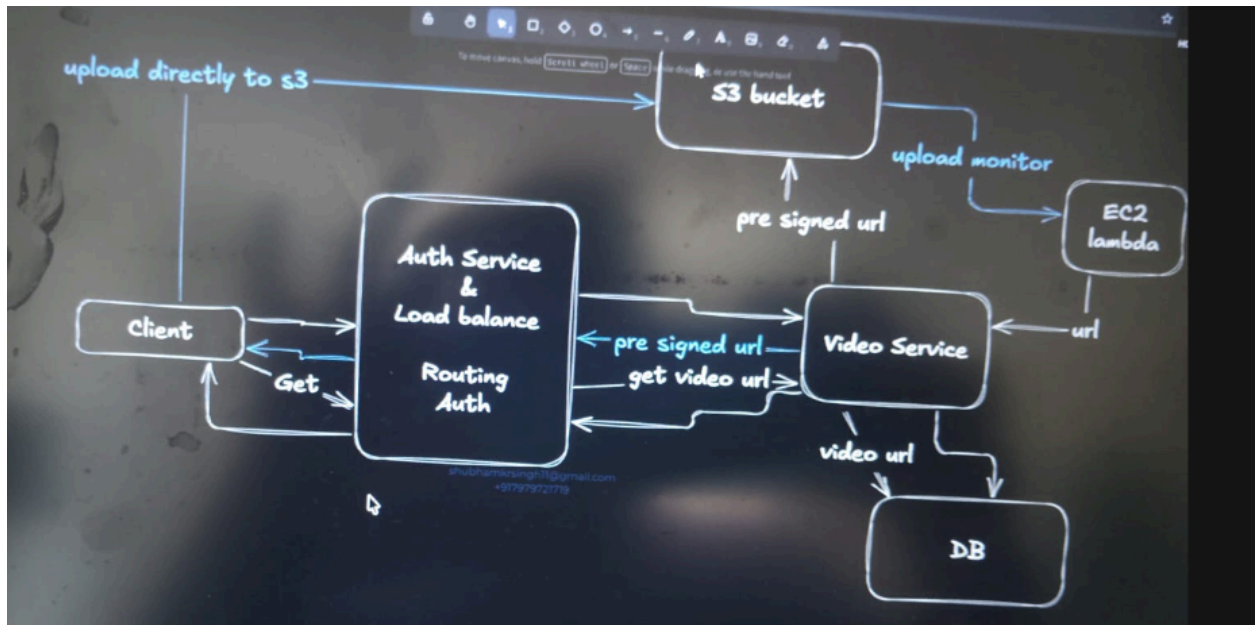


We use a pre-signed URL, which is a temporary URL that allows the client to upload the video directly to S3.

3. Now once the video gets uploaded, how does the client know that the upload is complete?







We use an upload event monitor. Once the upload is finished in S3, it triggers an event.

That event notifies our EC2 server, which then informs the Video Service.

The Video Service stores the video URL in the database and updates the status.

And then we create one fetch API that gets the video URL from the database.

4. Each upload service has its own notification service. S3 already has its own notification mechanism.
So when we upload a video, we don't need a separate upload monitor service.

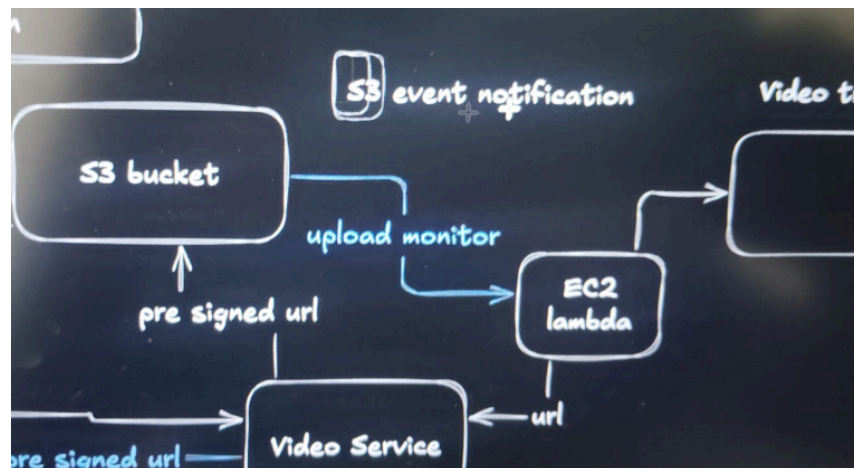
Why Upload Monitor Is Not Needed

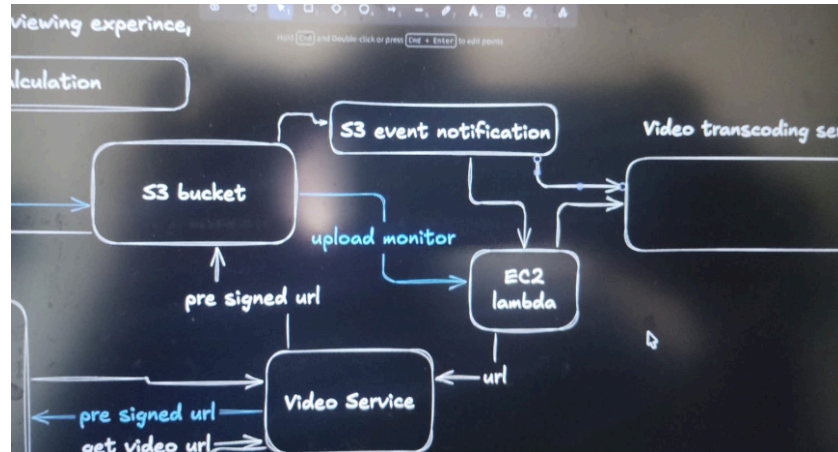
Amazon S3 already provides **Event Notifications**.

When a file is successfully uploaded, S3 can automatically trigger:

- AWS Lambda
- SQS
- SNS
- Or send event to an HTTP endpoint

So instead of building a separate “upload monitor” service, we rely on S3's built-in event system.





5. How Do We Serve Video in VOD?

To serve videos efficiently, we use a transcoding service like FFmpeg to convert uploaded videos into multiple resolutions and generate small chunks(manifest file). This enables adaptive bitrate video, where the user dynamically adjusts quality based on network speed. The processed chunks are stored in S3 and delivered via CDN.

What is CDN?

A **Content Delivery Network (CDN)** is a distributed network of servers placed in multiple geographic locations.

Example:

Amazon CloudFront

Cloudflare

What Problem Does CDN Solve?

Imagine:

- Your S3 bucket is in Mumbai.
- The user is in USA.

Without CDN:

- Request travels USA → India → USA.
- High latency
- More buffering

With CDN:

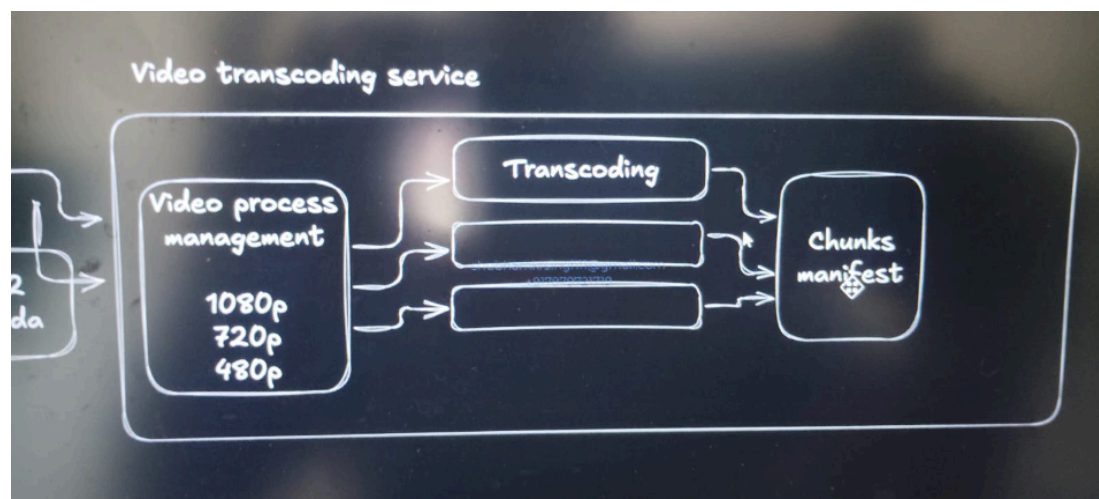
Video is cached in nearby edge location (USA).

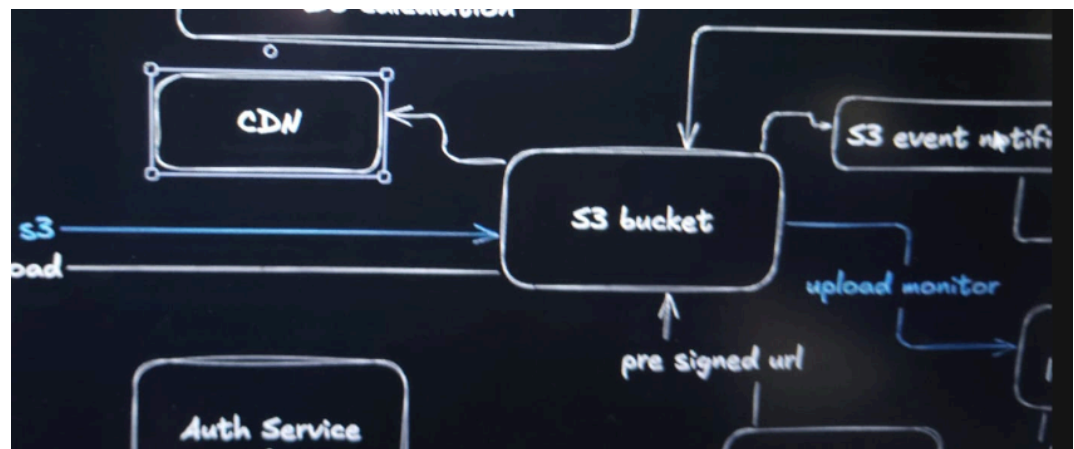
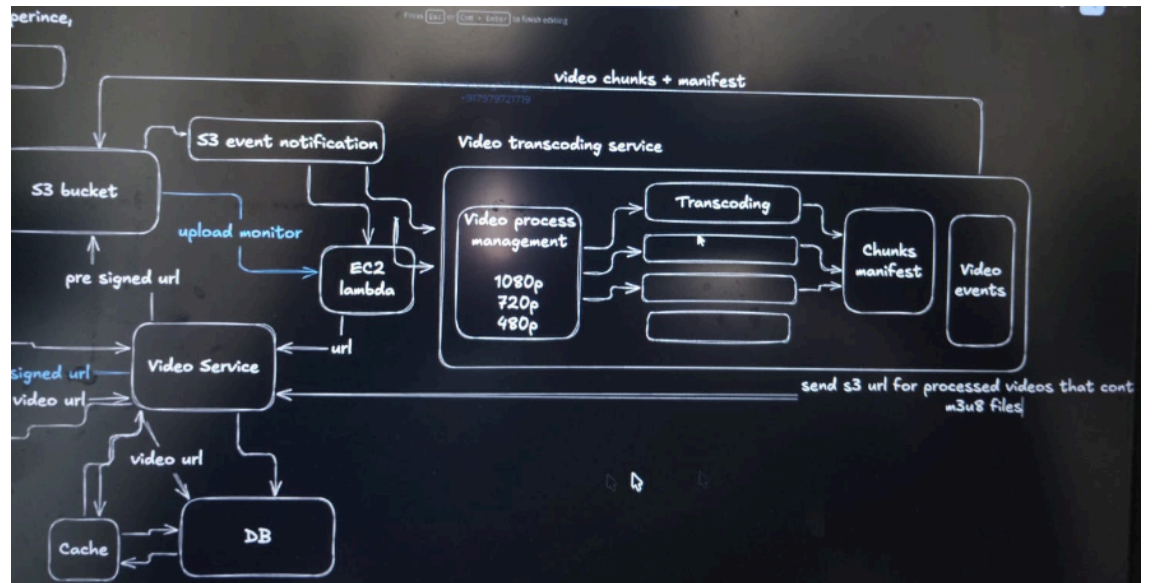
- Faster response
- Less buffering
- Better streaming quality

For Video Streaming Specifically

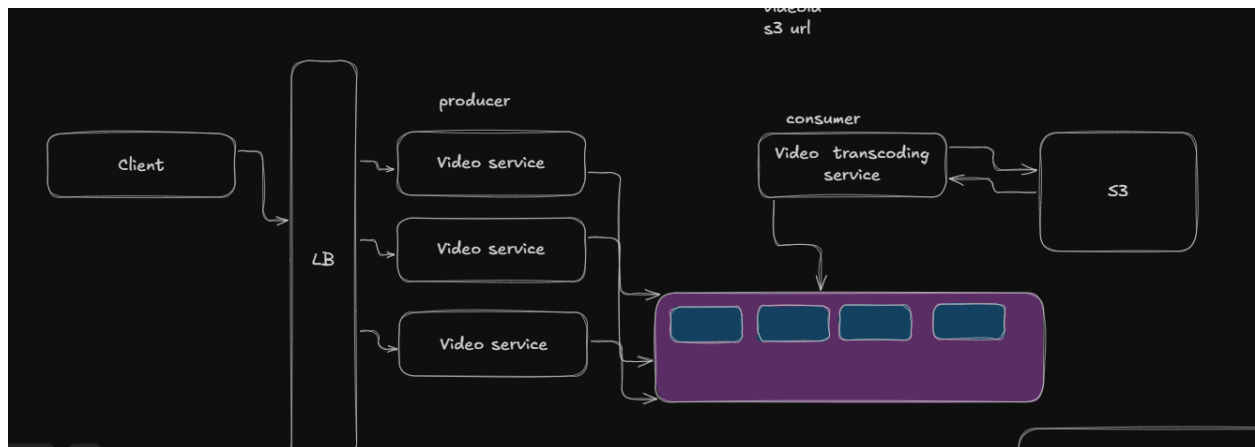
CDN helps with:

- Faster chunk delivery (.ts files)
- Reduced buffering
- Better startup time
- Handles millions of users
- Reduces load on S3

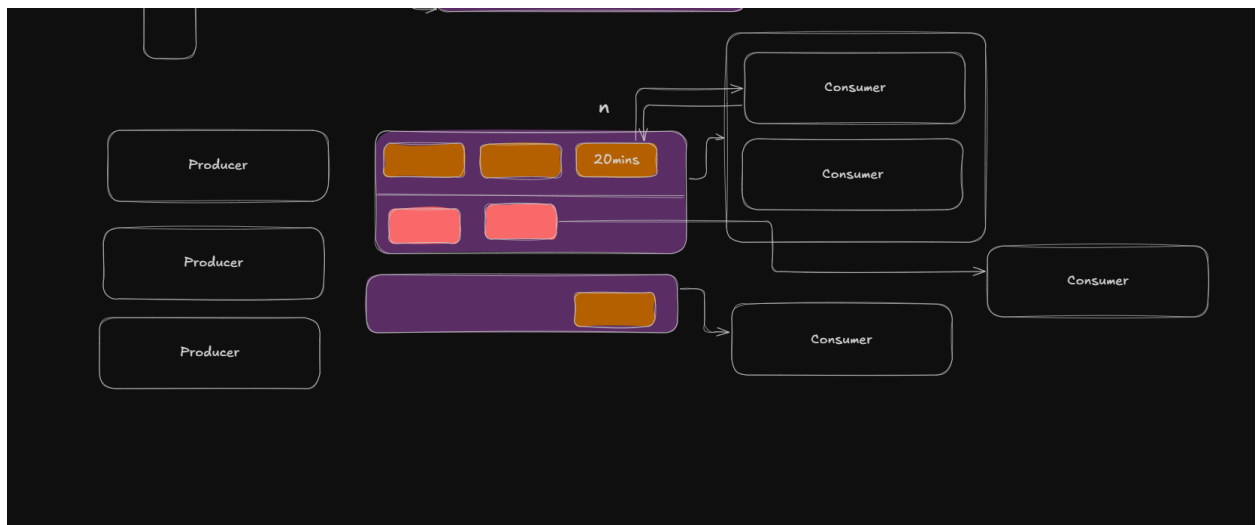




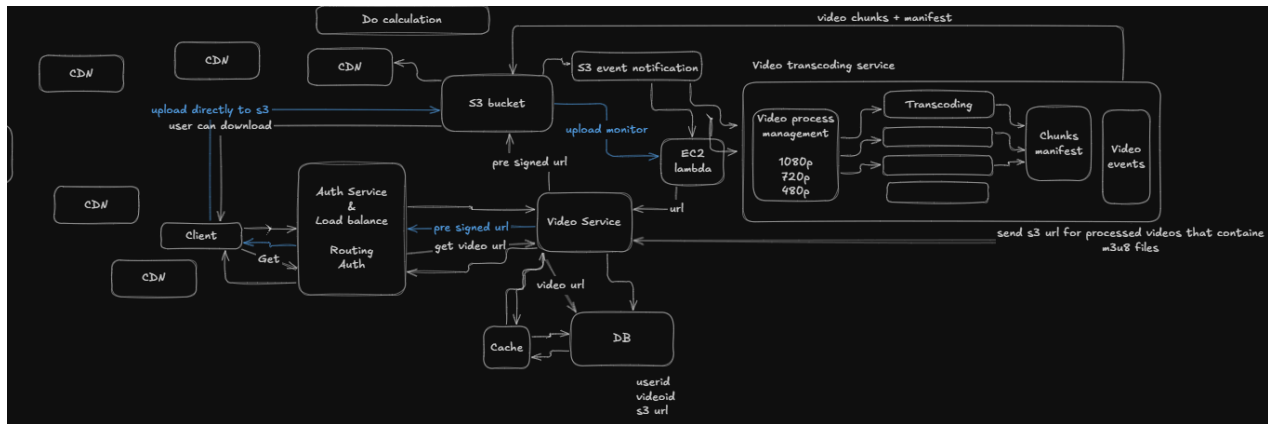
6. What problem in video transcoding service?
Queue service should be manager



7. More video in Queue
- We have to scale queue



8. Full System Design of Video Solution.



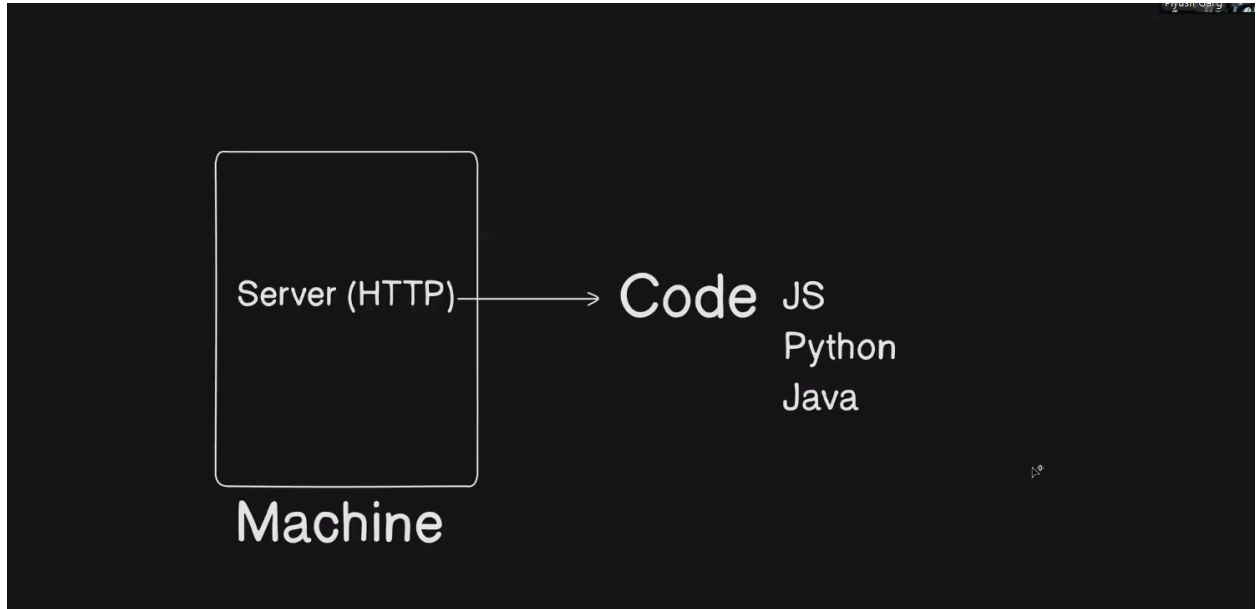
Homework:

Read about queue infrastructure (Full details of queue)

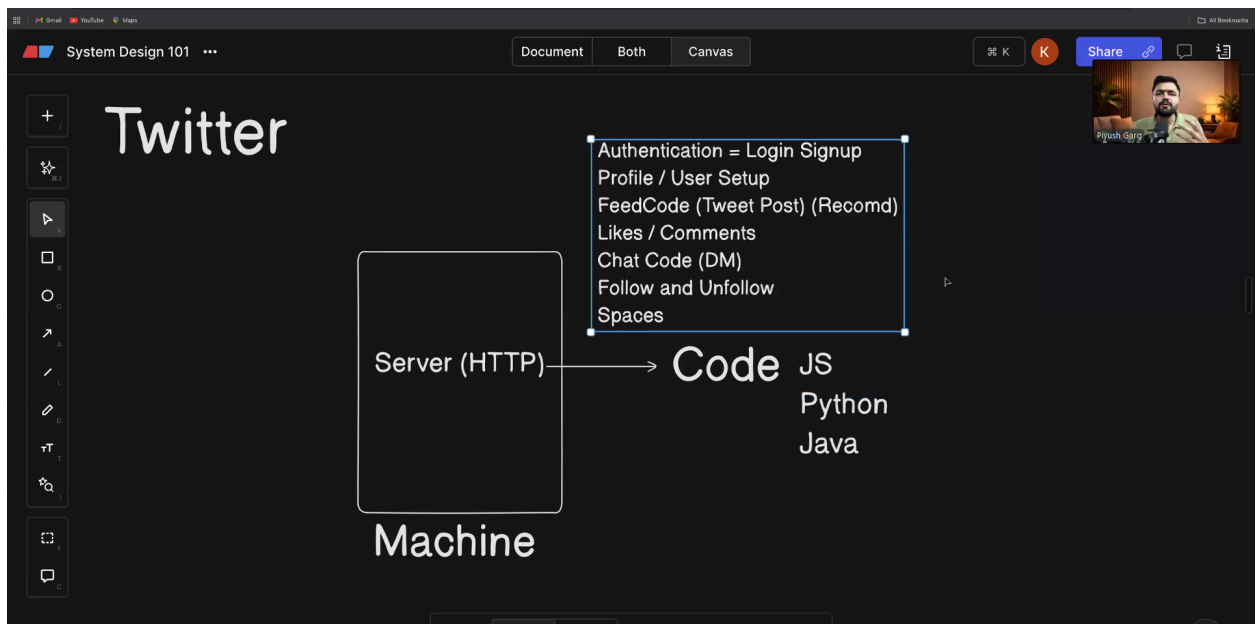
Monolith and microservice

1. Let's understand monolith

Let's take machine and machine ka ander ak HTTP server run krh rha hai and at the end server means backend code

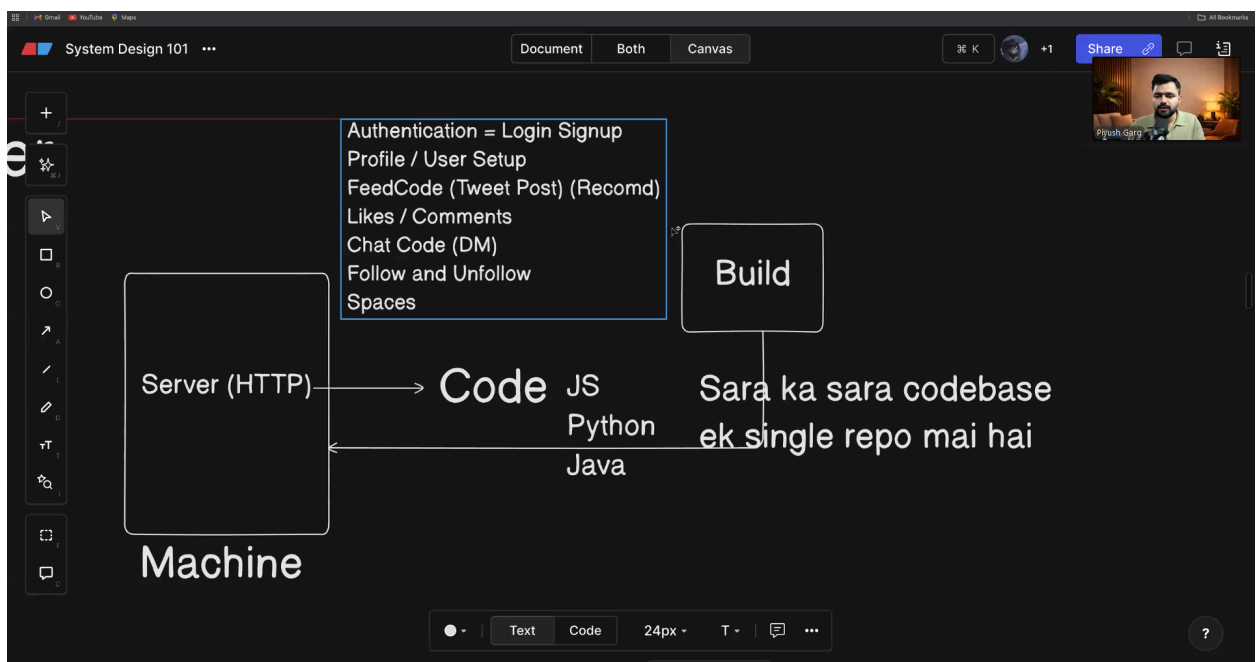


To Understand lets build Twitter

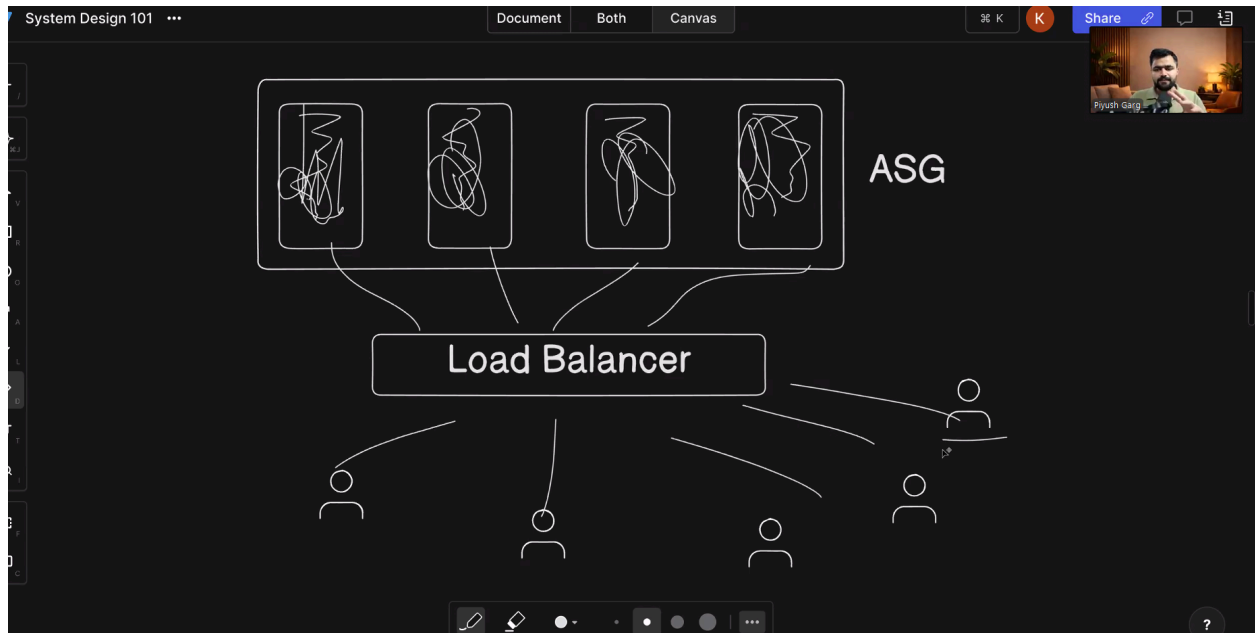


Overtime code get grow
All code in single repo
Build get deploy on AWS server

Sara ka sara codebase ek single repo mai hai



Now let's do horizontal scaling so each server have same number of code according on monolith architecture



Monolith

Sara ka sara code
ek single repo mai hau

Now user complain about our service

The word "Monolith" is displayed in a white, sans-serif font. The letter "M" is enclosed within a white rectangular border with rounded corners. The background is a solid dark grey or black.

Monolith

Sara ka sara code
ek single repo mai hau